TEAM-FLUX

Hyperlocal Delivery Intelligence Platform

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1.Problem Statement

Hyperlocal delivery platforms face continuous operational challenges caused by fluctuating customer demand, traffic congestion, adverse weather, festival surges, rider availability, and inventory shortages. These factors directly impact delivery times, customer satisfaction, operational costs, and revenue.

Most raw operational datasets remain fragmented across multiple sources and are not structured for multidimensional analysis or predictive modeling. Decision-makers require an integrated data warehouse and interactive Business Intelligence (BI) system to monitor real-time performance, analyze historical trends, identify bottlenecks, and proactively optimize logistics operations.

This project aims to design and implement an end-to-end analytics platform that transforms raw supply-chain and customer-experience data into actionable insights through data warehousing, data mining, and BI visualization.

2. Data Description

The project utilizes two real-world datasets:

📦 **Supply Chain Dataset**

Contains operational attributes such as:

- Delivery time and distance
- City and route information
- Traffic and weather conditions
- Vehicle type
- Festival indicators
- Restaurant/store details
- Order timestamps

⭐ **Fast Delivery Agent Reviews Dataset**

Contains customer-experience features such as:

- Delivery agent name
- Customer ratings

- Review text
- Order accuracy
- Product availability
- Discounts and price ranges
- Location

These datasets will be integrated through common dimensions such as time, city, and delivery agents to construct a unified analytical warehouse.

3. Objectives

The primary objectives of the project are:

1. Design a star-schema-based data warehouse for hyperlocal delivery operations.
2. Develop an ETL pipeline in R to clean, transform, and integrate raw datasets.
3. Perform OLAP analysis to study delivery performance across cities, time periods, traffic levels, and weather conditions.
4. Build predictive models to classify delayed versus on-time deliveries.
5. Segment delivery agents, customers, and cities using clustering techniques.
6. Discover service-quality and operational patterns using association-rule mining.
7. Detect anomalous orders and suspicious delivery records through outlier analysis.
8. Develop interactive Power BI dashboards for executive-level decision support.

4. Technologies & Tools Used

Layer	Tools
Data Processing & ETL	R, tidyverse, dplyr, tidyr
Data Warehouse	SQLite / PostgreSQL
Data Mining & ML	caret, e1071, arules, cluster
Text Mining	tidytext, tm
OLAP & BI	Power BI
Visualization	ggplot2, plotly
Version Control	GitHub
Documentation	Markdown / LaTeX / Word

5. Expected Outcomes

By the end of the project, the system is expected to deliver:

- A fully integrated supply-chain data warehouse supporting multi-dimensional analysis.
- Power BI dashboards enabling drill-down and slice-and-dice operations across business dimensions.
- Accurate classification models for delivery-delay prediction and customer-rating analysis.
- Clearly defined operational clusters for agents, customers, and cities.
- Actionable association rules revealing common service failures and demand patterns.
- Automated anomaly detection reports highlighting suspicious or extreme delivery cases.
- Business recommendations for improving logistics efficiency, staffing during peak hours, and inventory planning.

CONCLUSION

This project studies how delivery companies can work faster and smarter by analyzing their past orders, routes, traffic, weather, and customer feedback. We combine all this information into one system and use it to find patterns, predict delays, and spot problems early. Interactive dashboards show city-wise performance and customer satisfaction so managers can quickly understand what is going well and what needs improvement. The goal is to help companies reduce delays, save costs, and improve customer experience.